

4DRE-VIO: 4D Radar-Enhanced Monocular Visual Inertial Odometry for Urban Environments

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Abstract—Localization is crucial for Intelligent Transportation Systems (ITS) as it provides the precise position and orientation necessary for the operation of autonomous vehicles. Monocular vision-inertial odometry (VIO) has gained extensive application due to its high accuracy and cost-effectiveness. In this paper, we propose 4DRE-VIO, a novel 4D radar-enhanced monocular VIO system that achieves accurate and cost-effective pose estimation for autonomous vehicles in urban environments. By tightly integrating 4D radar measurements, we address the issues of scale ambiguity, scale drift, and dynamic object interference associated with monocular VIO. Specifically, we integrate 4D radar Doppler velocity with non-holonomic constraints (NHC) to provide a reliable velocity observation. Moreover, this velocity is directly used to construct a monocular scale observation, ensuring stable and accurate scale estimation. To mitigate the impact of dynamic objects on VIO, which is based on static environment assumptions, we introduce a 4D radar-enhanced visual semantic segmentation algorithm for dynamic object recognition. Additionally, we utilize a factor graph to tightly integrate measurements from the 4D radar, IMU, and monocular camera. The design of adjacent factors and optimal co-visible factors ensures the efficient use of visual information. An adaptive fusion strategy based on driving conditions maximizes the benefits of each sensor. We validate the effectiveness of the proposed method through extensive testing on a large dataset collected in urban environments, including open streets and residential districts. Compared to typical VIO and 4D radar-related odometry, our method demonstrates superior performance on our dataset and the public dataset.

Index Terms—Enhanced monocular visual inertial odometry, multi-sensor fusion, autonomous vehicle positioning, monocular scale estimation, dynamic object recognition.

I. INTRODUCTION

LOCALIZATION technology has an important influence on the safe and efficient operation of Intelligent Transportation Systems (ITS) [1]. Particularly for autonomous vehicles, which have received considerable attention in recent years, continuous, reliable, and accurate localization in urban environments is essential [2], [3]. This is a key input for

downstream tasks like trajectory prediction, motion planning, and control [4]. The Global Navigation Satellite System (GNSS) is a widely adopted positioning solution [5], that offers significant advantages such as high accuracy, compact size, and moderate cost, making it an indispensable sensor for autonomous vehicles. However, in urban environments, GNSS is susceptible to multipath effects and non-line-of-sight reception [6], [7]. This can lead to GNSS receivers receiving delayed or wrong data, adversely affecting positioning [8]. To improve this situation, [8] achieves the transmission of GNSS information to the autonomous vehicle with the highest accuracy and efficiency through data compression techniques, which is very beneficial for improving the positioning performance. Fault tolerance and data omission are also supported [8]. On the other hand, leveraging other sensors for reliable auxiliary support is also recognized as a popular solution. Odometry systems based on vision [9], [10], LiDAR [11], and Inertial Measurement Units (IMUs) have demonstrated excellent performance in urban environments, making them reliable solutions. Combining different sensors such as LiDAR, ultrasonic ranger finder [12], and wheel speed encoder [13] has achieved a better ability of the object detection [12] and positioning capabilities for autonomous vehicles. Among these, monocular visual inertial odometry (VIO) has garnered significant attention from researchers due to its high accuracy and cost-effectiveness [14].

Unfortunately, vision-based odometry is significantly affected by lighting conditions and requires static scenes and sufficient environmental characteristics to accurately estimate poses. Additionally, monocular visual odometry (VO) faces challenges such as scale ambiguity and scale drift. The monocular scale cannot be obtained directly and gradually drifts with scene changes [15]. Integrating with the IMU can help recover the scale. However, during constant velocity or constant turning of the vehicle, the monocular scale becomes unobservable due to insufficient acceleration excitation. To address this issue, many scholars adopt multi-sensor fusion solutions, such as incorporating LiDAR [16], [17]. Although these methods have demonstrated excellent performance, LiDAR remains unaffordable for many applications and is affected by challenging weather conditions. In contrast, the recently launched 4D radar is more cost-effectiveness and stable in harsh environments, including rain, snow, fog, and difficult lighting conditions. Its excellent performance has attracted widespread attention from academia and industry.

With advantages that extend beyond cameras, 4D radar serves as a strong complement to vision and inertial sensors.

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Firstly, compared to cameras, the spatial information provided by 4D radar can resolve the scale uncertainty issue in monocular VO. Secondly, the scene motion information from 4D radar can help mitigate the adverse effect of dynamic objects on odometry. However, the point clouds from 4D radar are sparse and noisy, resulting in poor odometry accuracy, particularly in attitude estimation. In contrast, observations from vision and IMU provide better attitude constraints. Additionally, 4D radar, monocular cameras, and IMUs are cost-effective. Therefore, exploring tightly coupled positioning methods using 4D radar, monocular cameras, and IMUs holds significant value for both academic research and industrial applications.

The integration of 4D radar, IMU, and monocular camera to achieve accurate and reliable pose estimation faces three key challenges. Firstly, as an emerging sensor, 4D radar has limited related research in the field of positioning. Compared to LiDAR point clouds, 4D radar point clouds are sparse and noisy, making it difficult to directly apply classical LiDAR and VIO methods. Therefore, it is necessary to consider the characteristics of 4D radar to design appropriate fusion methods. Secondly, pixels associated with dynamic objects in images may lead to incorrect pose estimation because they violate the static environment assumption. This problem is more pronounced on open streets with many dynamic objects. Finally, designing suitable fusion strategies for different driving conditions to fully exploit the advantages of each sensor to achieve accurate and reliable pose estimation is a challenge.

To address these challenges, this study proposes a novel 4D radar-enhanced monocular visual inertial odometry, 4DRE-VIO. This system is a tightly coupled solution based on a factor graph. Firstly, considering that the noisy and sparse characteristics of the 4D radar point clouds lead to poor matching accuracy based on the point cloud, we use Doppler velocity information to infer ego velocity. Combined with the non-holonomic constraints (NHC), we develop a reliable velocity observation. Meanwhile, we utilize the velocity to construct a direct monocular scale observation, ensuring stable and accurate scale estimation. Next, to mitigate the disruption caused by dynamic objects to the monocular VO, we propose a dynamic object detection algorithm based on 4D radar-enhanced visual semantic segmentation. Additionally, we construct relative pose measurements in the visual sliding window, which cleverly and flexibly integrates the visual constraints of the sliding window into the factor graph. Finally, 4D radar, IMU, and vision measurements are tightly integrated based on the factor graph with a sliding window. Considering the characteristics of static and turning conditions, the fusion strategy is adaptively adjusted to fully leverage the advantages of each sensor, achieving accurate and consistent pose estimation in urban environments. The effectiveness of the proposed method is confirmed by extensive experiments on the public dataset NTU4DRadLM [18] and our dataset. We highlight the contributions of this paper as follows:

- We propose a novel factor graph-based 4D radar-enhanced monocular visual inertial odometry framework, 4DRE-VIO, which leverages the unique spatial and velocity information provided by 4D radar to address the scale ambiguity and dynamic object interference issues

associated with monocular VIO. Furthermore, our framework can be easily extended to integrate other multi-source sensors.

- In the proposed framework, we construct a direct monocular scale observation model based on 4D radar Doppler velocity information and incorporate scale optimization within the factor graph framework. This provides a new theoretical approach to address the scale unobservability problem in monocular VIO under vehicle degradation motion.
- We propose a 4D radar-enhanced visual semantic segmentation algorithm for dynamic object detection, which utilizes 4D radar point cloud and Doppler information to improve dynamic object recognition. This provides a novel technical approach to effectively mitigate dynamic object interference on VIO.
- We propose two key technical innovations in factor graph design: the design of visual adjacent factors and optimal co-visible factors, as well as an adaptive fusion strategy. The former ingeniously introduces visual constraints from the sliding window into the factor graph, enhancing system consistency and flexibility. The latter achieves adaptive fusion of visual and IMU attitude constraints by dynamically adjusting the covariance of visual factors, improving the accuracy of yaw angle estimation.

The rest of the paper is organized as follows. Section II introduces the related work. Section III describes the overall structure of the proposed 4DRE-VIO. Section IV introduces ego velocity estimation and monocular VO, collectively referred to as the multi-sensor front-end. Section V details the structure of the factor graph. Section VI provides experimental validation and discussion of the proposed algorithm. Finally, the conclusion is presented in Section VII.

II. RELATED WORK

A. Visual Inertial Odometry

VIO has garnered widespread attention from researchers due to its high accuracy and cost-effectiveness, which primarily consists of traditional VIO and learning-based VIO. Depending on the fusion approach used, traditional VIO can be divided into filter-based and optimization-based methods. Filter-based VIO is implemented in two steps: first, predicting the state using the IMU, and then updating the state based on visual observations. The Multi-State Constraint Kalman Filter (MSCKF) [19] is the most classic filter-based VIO framework. MSCKF incorporates camera poses at different times into the state vector instead of visual feature points, and uses geometric constraints between multiple camera poses to construct the observation model to update the system state [19]. MSCKF 2.0 [20] employs the First Estimated Jacobian method to improve its consistency. Subsequently, many excellent works have emerged to further improve and refine MSCKF [21], [22], [23], [24].

Optimization-based VIO estimates pose states by solving nonlinear least squares problems, which has been proven to achieve better performance [25]. OKVIS [26] uses a keyframe-based sliding window optimization method for state

estimation. To mitigate the impact of dynamic objects such as people and clouds, OKVIS2 [27] introduces a semantic segmentation network. VINS-Mono [28], VINS-Fusion [29], [30] and ORB-SLAM3 [31] are currently popular VIO systems with excellent performance. However, during initialization they face the problem of being unable to observe the scale via IMU data for short periods, which causes poor performance. This is the main reason why stereo-inertial methods are superior to monocular-inertial methods. To address this issue, VI-DSO [32] introduces IMU measurements and a dynamic marginalization strategy based on DSO [33], which allows the system to initialize at any scale. However, large scale initiation errors can cause the algorithm to fail in large outdoor environments [34]. On this basis, they propose DM-VIO [34], which employs a delayed marginalization strategy to ensure that the initialization process captures the full photometric uncertainty and improves monocular scale estimation. Nevertheless, scale drift in large-scale environments and the unobservable scale under vehicle degradation motion remain major challenges to accurate pose estimation in monocular VIO.

With the advancement of deep learning technologies, learning-based VIO has garnered significant attention from researchers in recent years. End-to-end learning-based VIO encodes and fuses image and IMU information using networks, trained through either supervised [35] or unsupervised [36] methods. iSLAM [37] integrates learning-based VO and IMU modules with graph optimization, enhancing performance through self-supervised learning. Adaptive VIO [38] uses non-linear optimization to construct the loss function, introducing a feedback mechanism that enables continuous online learning. This type of learning-based VIO combines the advantages of geometric theory and deep learning, demonstrating superior performance compared to end-to-end VIO in various scenarios [39]. However, the computational resource demands of deep learning present challenges for real-time deployment [40]. Additionally, there are shortcomings in generalization performance and data dependency [40]. Consequently, traditional VIO remains widely used in practical applications [39].

B. Radar-Related Visual/Inertial Odometry

Millimeter-wave radar odometry (RO) methods have garnered increasing attention from scholars, mainly due to their weather resilience [41], [42], [43]. Compared to traditional radar, 4D radar provides a 3D point cloud of the scene, containing more abundant information. This opens new opportunities for achieving reliable all-weather 3D positioning. Current research tends to focus on positioning based on 4D radar-only, or fusing 4D radar with IMU. The noisy and sparse characteristics of 4D radar point clouds result in poor accuracy when directly using point cloud registration methods. Reference [44] improves the accuracy of point cloud registration-based odometry by calculating the probability distribution of each point. Reference [45] proposes using 4D radar Doppler velocity to estimate ego velocity, and then introducing a velocity pre-integration factor in the pose graph optimization to achieve more accurate pose estimation. However, the

accuracy of 4D radar-only positioning is limited, especially in large-scale scenarios. References [46], [47], and [48] combines IMU with 4D radar-based ego velocity estimation further improves positioning accuracy. In geometrically degraded environments, these methods can fail. In addition, only the velocity observation cannot effectively suppress the drift of IMU integration, especially the attitude drift.

To this end, in the framework of ROVIO [49], [50] integrates 4D radar, IMU, and thermal images to improve the stability of the algorithm under challenging visual conditions. However, this is a loosely coupled approach that only directly fuses the results of VO and RO in the back-end. The data from 4D radar does not directly compensate for the shortcomings of VO to improve its performance. In the framework of ORB-SLAM [39], [51] introduces 4D radar measurements at different stages of VO, achieving better performance than the loosely coupled approach. However, this method relies primarily on visual data for rotation estimation [39], which may lead to reduced positioning performance when VO performance is poor and the output switches to RO. To mitigate this issue, further integration of IMU data could improve the accuracy of short-term attitude estimation [39]. However, research on the integration of 4D radar, monocular cameras, and IMU for positioning remains limited. To fill this gap, we aim to develop an efficient and reliable tightly coupled fusion framework using all three sensors. We will not only integrate 4D radar measurements at multiple stages of the VO, but also tightly integrate IMU measurements to improve attitude accuracy, leveraging the advantages of each sensor for high-precision positioning in urban environments.

III. SYSTEM OVERVIEW

This study introduces 4DRE-VIO, a novel 4D radar-enhanced monocular visual inertial odometry, which achieves tight coupling based on factor graph optimization. The overall architecture of the proposed 4DRE-VIO is illustrated in Fig. 1. The system receives input from 4D radar, IMU and a monocular camera. Firstly, due to the noisy and sparse nature of the 4D radar point clouds, we do not use the traditional point cloud registration method to obtain relative measurements. Instead, we primarily use the Doppler velocity measurements. Specifically, in the ego velocity estimation module, we use the RANSAC algorithm to perform dynamic-static segmentation on the 4D radar point cloud and then estimate the ego velocity from the static segmented point cloud. This significantly benefits monocular scale recovery and positioning accuracy. Secondly, zero velocity detection, based on IMU acceleration information and ego velocity estimation results, is performed to provide markers for processing static conditions. In addition, a monocular VO module is developed based on DM-VIO [34]. Note that we do not enable the IMU module in DM-VIO, making the algorithm a purely monocular VO-DSO [33]. We make three key improvements: 1) A dynamic object recognition module using 4D radar-enhanced visual semantic segmentation is proposed, which mitigates the disruption of dynamic objects to the monocular VO module based on static assumptions. 2) In order to flexibly integrate visual constraints from the sliding window into the

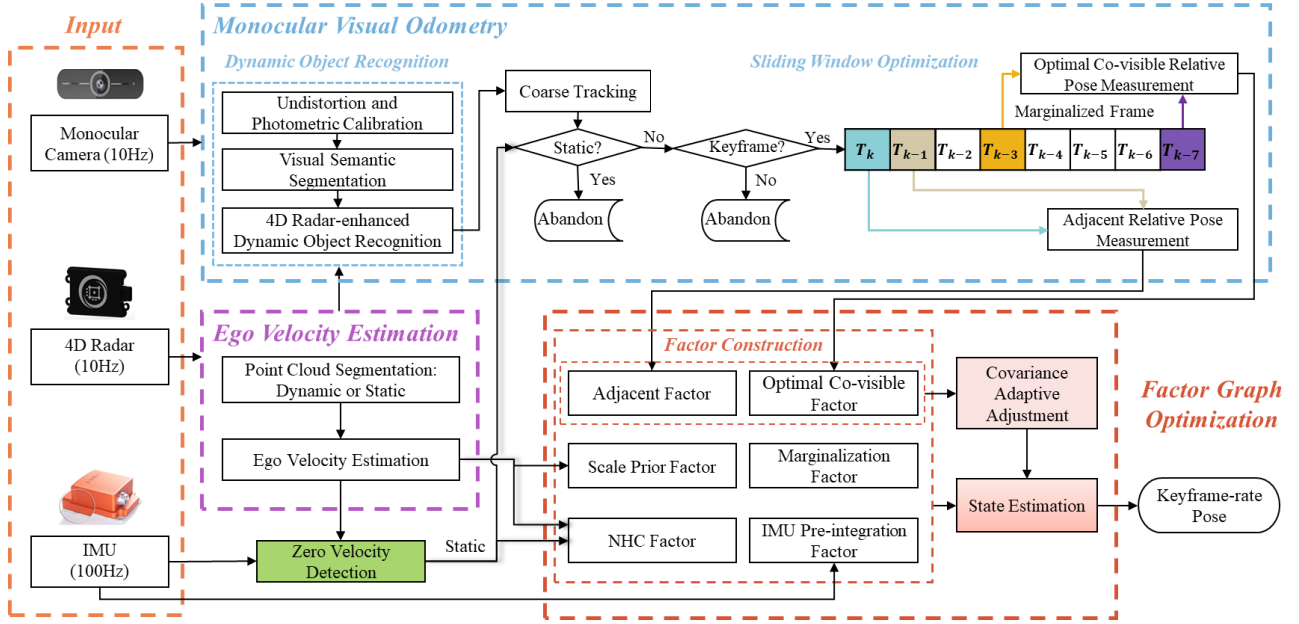


Fig. 1. Overall architecture of the proposed 4DRE-VIO.

factor graph, we construct adjacent relative pose measurement and optimal co-visible relative pose measurement, respectively. 3) Using the zero velocity detection results, we design a strategy to avoid adding keyframes to the sliding window during static conditions. This helps mitigate the disruption of good spatial distribution within the sliding window caused by the vehicle's static state. These processes will be detailed in Section IV.

Finally, the visual adjacent factors, optimal co-visible factors, scale prior factors, NHC factors, IMU factors, and marginalization factors are jointly added to the factor graph to perform pose graph optimization. Among them, we design an covariance adaptive adjustment strategy for the visual factors, which significantly improves the accuracy of yaw angle estimation. The NHC factor is a velocity observation constructed by combining the ego velocity estimation results with the NHC. It also provides good zero velocity constraints when the vehicle is static. Detailed information about this module will be provided in Section V.

Here are the definitions of the notation used in this paper. $(\cdot)^w$ represents the world coordinate system, with its Z-axis opposite to the direction of gravity. $(\cdot)^c$ represents the camera coordinate system, and $(\cdot)^r$ represents the 4D radar coordinate system. $(\cdot)^b$ represents the body coordinate system, which is the same as the IMU coordinate system. $(\cdot)^{ci}$ represents the i -th keyframe in camera frame, and $(\cdot)^{bi}$ represents the i -th body frame. We use the rotation matrix $\mathbf{R}_b^w$ and the Hamilton quaternion $\mathbf{q}_b^w$ to denote the rotation from frame $(\cdot)^b$ with respect to frame $(\cdot)^w$. $\mathbf{p}_b^w$ denotes the translation vector of $(\cdot)^b$ to $(\cdot)^w$. We also use $\mathbf{T}_b^w \in SE(3)$ to describe the relative pose transformation of $(\cdot)^b$ with respect to $(\cdot)^w$, which includes both the translation $\mathbf{p}_b^w$ and rotation $\mathbf{R}_b^w$.

IV. MULTI-SENSOR FRONT-END

To achieve reliable and accurate state estimation by tightly coupling the 4D radar, IMU and monocular camera

measurements in the factor graph, it is essential to finely process the information from each sensor at the front end. In this section, we detail the ego velocity estimation module based on 4D radar, the zero velocity detection module based on IMU, and the monocular VO module. In addition, to mitigate the impact of dynamic objects on the monocular VO module, we propose a 4D radar-enhanced visual dynamic object recognition algorithm.

A. Ego Velocity Estimation

4D radar not only provides 3D measurements of the environment, but also provides Doppler velocity measurements for each point in the scene. This provides valuable information for addressing issues of monocular scale uncertainty and the removal of dynamic objects. To fully utilize this information, we introduce a RANSAC-based method for dynamic-static point cloud segmentation and ego velocity estimation.

4D radar detects the position and motion information of a target by transmitting radio waves and receiving their reflections. For a given target point P , the measurement obtained by 4D radar includes: 3D position $\mathbf{p}^r = [x_p^r \ y_p^r \ z_p^r]^T$, and Doppler velocity measurement v_r . v_r represents the relative velocity between the target point and the radar along the radial direction with the formula:

$$-v_r = \frac{\mathbf{p}^r}{\|\mathbf{p}^r\|} \cdot (\mathbf{v}_{ego}^r - \mathbf{v}_{obj}^r) = \mathbf{d}_p^r \cdot (\mathbf{v}_{ego}^r - \mathbf{v}_{obj}^r) \quad (1)$$

where $\mathbf{d}_p^r = \begin{bmatrix} x_p^r/l \\ y_p^r/l \\ z_p^r/l \end{bmatrix}$, $l = \sqrt{x_p^{r2} + y_p^{r2} + z_p^{r2}}$. $\mathbf{v}_{obj}^r$ represents the velocity of the measurement point. $\mathbf{v}_{ego}^r = [v_{ego_x}^r \ v_{ego_y}^r \ v_{ego_z}^r]^T$ represents the velocity of the vehicle in 4D radar frame. From (1), it is evident that both $\mathbf{v}_{ego}^r$ and $\mathbf{v}_{obj}^r$ are unknown. Given that static target point have $\mathbf{v}_{obj}^r = \mathbf{0}$. Therefore, it is more advantageous to utilize solely static points for ego velocity estimation.

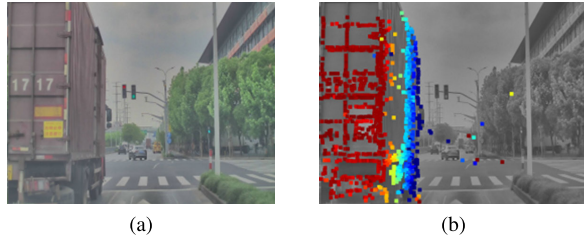


Fig. 2. Example of a dynamic scene. (a) Raw image. (b) The result of the projection of the high-gradient points selected by DSO back onto the raw image.

Therefore, we use the RANSAC [52] to simultaneously solve the problems of 4D radar point cloud dynamic-static segmentation and ego velocity estimation. We set outliers as dynamic points and inliers as static points, where the parameter to be estimated is the ego velocity. For a frame of the 4D radar point cloud, we construct a set of (2) based on the Doppler velocity measurement v_r of each extracted static point and then estimate the ego velocity $\mathbf{v}_{ego2D}^r = [v_{ego_x}^r \ v_{ego_y}^r]^T$ using the least squares algorithm. This is because we are primarily considering the vehicle in planar motion.

$$\begin{bmatrix} \frac{x_p^r}{l} & \frac{y_p^r}{l} \end{bmatrix} \mathbf{v}_{ego2D}^r = -v_r \quad (2)$$

In addition, in each iteration, we set up to use 2 static point measurements for least squares estimation. Therefore, the solution using linear least squares is as follows:

$$\mathbf{v}_{ego2D}^r = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \vec{v}_r$$

$$\mathbf{H} = \begin{bmatrix} \frac{x_{p1}^r}{l_1} & \frac{y_{p1}^r}{l_1} \\ \frac{x_{p2}^r}{l_2} & \frac{y_{p2}^r}{l_2} \end{bmatrix}, \quad \vec{v}_r = \begin{bmatrix} -v_{r1} \\ -v_{r2} \end{bmatrix} \quad (3)$$

The covariance of the ego velocity estimate can be calculated as follows:

$$Cov(\mathbf{v}_{ego2D}^r) = (\mathbf{H}^T \mathbf{H})^{-1} \|\vec{v}_r - \mathbf{H} \mathbf{v}_{ego2D}^r\|^2 \quad (4)$$

B. Zero Velocity Detection

Zero velocity detection and update is a commonly used technique in inertial navigation [53]. It is used to mitigate the integration drift of IMU at vehicle static. For zero velocity detection, we adopt the commonly used method of acceleration variance detection. We also use 4D radar velocity estimation results for zero velocity re-validation, which helps to eliminate the interference caused by constant velocity linear motion. The specific strategy is the following: If the variance of the IMU three-axis acceleration between five consecutive time steps and the ego velocity are less than the set threshold, the vehicle is considered to be static.

C. Monocular Visual Odometry

Monocular VO uses the images as input to estimate the relative pose of the camera. This section will introduce our improvement strategies towards DSO [33]: the dynamic object recognition algorithm and the construction of visual relative pose measurements.

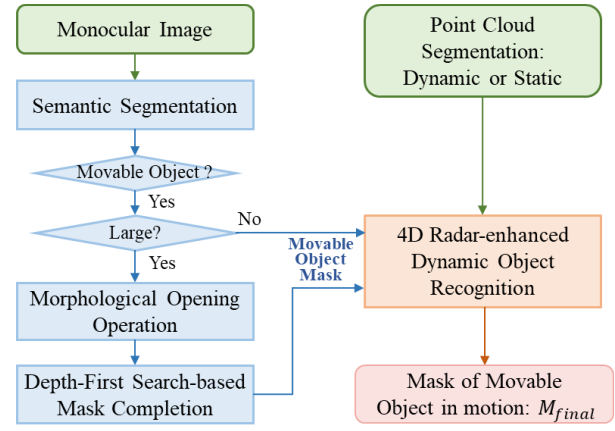


Fig. 3. Flow of dynamic object recognition.

1) *4D Radar-Enhanced Visual Dynamic Object Recognition*: Assuming a static scene is an important prerequisite for VO. Pixels in images associated with dynamic objects can lead to incorrect pose estimation, and this problem is more severe in urban environments with many dynamic objects. Fig. 2 shows a case where dynamic objects cause anomalies in VO pose estimation, primarily due to high-gradient points being concentrated on the dynamic objects. It can be seen that there is a moving truck on the left side of the ego vehicle. The high-gradient points extracted within the DSO-based visual sliding window are distributed mainly on the moving truck, resulting in an abnormal relative pose estimation. To avoid such problems, dynamic objects should be removed from the images. This ensures that the relative pose estimation is based only on static points. We propose a dynamic object recognition algorithm based on 4D radar-enhanced semantic segmentation. Dynamic object recognition is illustrated in Fig. 3. First, semantic segmentation is performed on monocular images to extract movable object masks (such as cars, trucks, and pedestrians, which are all movable objects). To deal with any abnormal segmentation results, we have developed a method for completing the mask to mitigate the problem of incorrect segmentation of large objects and to achieve accurate segmentation results. Then, by integrating static and dynamic point clouds from 4D radar, accurate moving state determination is achieved.

Firstly, to perform semantic segmentation on monocular images, we adopt the InternImage [54], which has the highest semantic segmentation accuracy on the Cityscapes dataset [55]. To obtain pixel-wise image semantic segmentation results, we feed the images directly into the InternImage model pre-trained on the Cityscapes dataset. InternImage performs well on most of the sequences even without any adjustment of the model parameters. However, when dynamic objects appear partially in the image, such as the large truck in Fig. 4a, the results are abnormal. This can cause dynamic objects to be mistaken for static objects, leading to incorrect relative pose estimation. To address this, when there is a large dynamic object, such as a large truck, we perform a mask completion operation on the segmentation mask of InternImage. We first perform pre-processing using the morphological opening operations to separate the incorrectly segmented

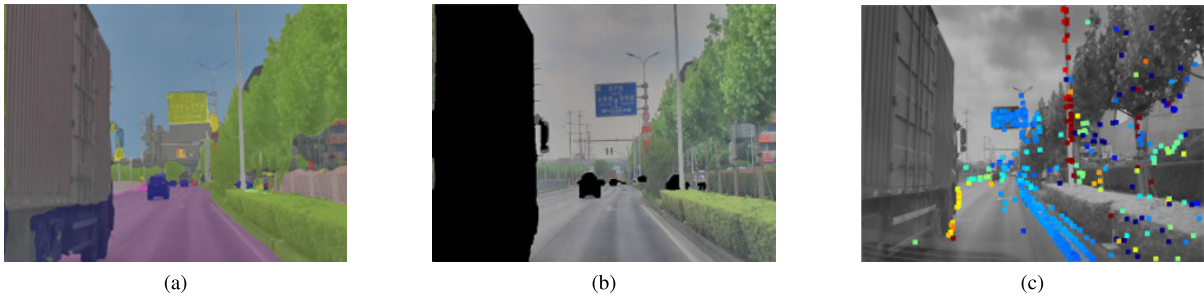


Fig. 4. Results of 4D Radar-enhanced Visual Dynamic Object Recognition. (a) An example showing the segmentation limitations of InternImage. The carriage part of the truck is incorrectly classified as a building and pole, and the light part is incorrectly classified as a traffic sign. (b) Result of our proposed mask complementing method, which obtains better segmentation results. (c) The result of projection selected high-gradient points in DSO back onto the raw image.



Fig. 5. Results of 4D radar dynamic-static point cloud segmentation to enhance dynamic object recognition. (a) The “movable object” mask with 4D radar dynamic-static point cloud projection results. Blue dots represent dynamic point clouds and green dots represent static point clouds. Note that there are many static point clouds on the mask of the operating vehicle parked on the side of the road, indicating a static target. (b) The “movable object in motion” mask.

pixels on the large truck from other pixels (e.g., buildings, etc.). Next, we use depth-first search for mask completion. These operations mitigate the problem of misclassification of large objects and allow us to obtain accurate “movable object” masks. The segmentation results in Fig. 4b demonstrate the effectiveness of the proposed mask completion method.

However, “movable object” is not necessarily in a state of movement, they can also be static. To accurately identify whether a movable object is in a state of movement or static, we propose a dynamic object recognition method with the help of 4D radar point clouds. The point clouds are projected onto the “movable object” mask and the state of movement or static is determined based on the number of dynamic and static point clouds on the mask. Finally, we keep only the accurate “movable object in motion” mask M_{final} . As shown in Fig. 5a and 5b, the proposed method provides accurate detection of dynamic objects.

With the generated M_{final} , we have improved the strategy for the selection of high-gradient points and the total photometric error function of the DSO. Specifically, when searching for high-gradient points in the specified gradient direction within each sub-block, M_{final} is used to determine whether the point is on a dynamic object. If it is, the point will be skipped and the search will continue to the next point. Although this ensures that the Harris points are static on the host frame, they can fall on dynamic objects when reprojected onto other keyframes within the sliding window, which are also called target frames. This is due to static objects getting occluded by dynamic objects. To further ensure that dynamic points are not involved in the construction of the photometric

error function within the sliding window, we check the reprojected target frame. These two improvements ensure that dynamic points are excluded from the pose estimation process, thereby reducing the disruption to monocular VO based on the static assumption. As shown in Fig. 4c, in the same dynamic scene as in Fig. 2, all the high-gradient points extracted by the proposed method fall on static objects, which is very beneficial for pose estimation.

2) *Visual Relative Pose Measurement Construction*: As shown in Fig. 6, the input images are divided into keyframes and non-keyframes. The keyframe pose is optimized several times in the sliding window until it is marginalized, called a marginalized frame. We only add keyframes to the sliding window and finally include them in the factor graph optimization. The length of the sliding window in DSO is 8, and we apply the same setting. To flexibly integrate visual observations within the sliding window into the factor graph, we propose two types of visual relative pose measurements. After each sliding window optimization, we calculate the relative transformation between the latest keyframe pose $\mathbf{T}_k$ and the second latest keyframe pose $\mathbf{T}_{k-1}$. This is the adjacent relative pose measurement. However, because $\mathbf{T}_k$ and $\mathbf{T}_{k-1}$ are optimized limited times within the sliding window, they are not as accurate as the oldest keyframe pose $\mathbf{T}_{k-7}$. Theoretically, the pose $\mathbf{T}_{margi}$ of marginalized frame in the sliding window has the highest accuracy. Therefore, we also calculate the relative transformation between $\mathbf{T}_{margi}$ and $\mathbf{T}_{k-7}$, which is referred as the optimal co-visible relative pose measurement. This long-term measurement improves the consistency of the pose estimation results, making the trajectory smoother. The calculation method is as follows:

$$\mathbf{T}_{margi}^{k-7} = (\mathbf{T}_{k-7})^{-1} \mathbf{T}_{margi} \quad (5)$$

V. FACTOR GRAPH OPTIMIZATION

Factor graphs are probabilistic graphical models that provide an intuitive and flexible representation of the joint probability distribution over variables, where nodes represent the state variables to be estimated, and factors encode constraints between these variables. The core objective of the factor graph optimization is to iteratively refine the state estimation by minimizing the error functions associated with the factors.

In this section, we present the factor graph optimization framework for tightly integrating 4D radar, IMU, and camera. The structure of the factor graph is shown in Fig. 6,

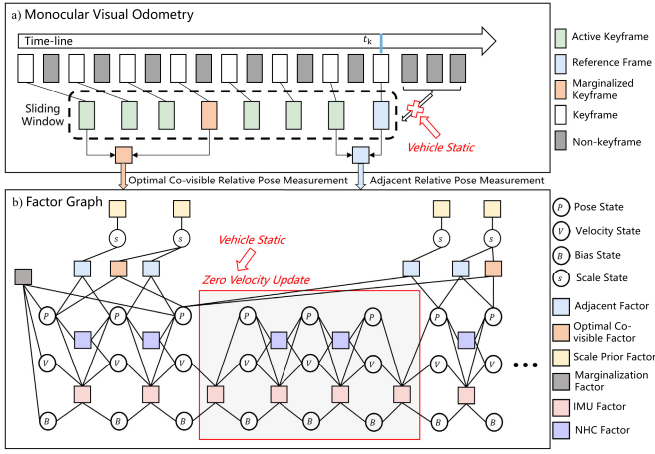


Fig. 6. Illustration of the factor graph of the 4DRE-VIO. a) Example of constructing adjacent relative pose measurement and optimal co-visible relative pose measurement in the monocular VO. b) Structure of the factor graph.

which includes visual adjacent and optimal co-visible factors, NHC factors, scale prior factors, IMU factors, and marginalization factors. When a new visual keyframe is created, a new state node is created in the factor graph, and then factor graph optimization is performed based on the sliding window. The state vector χ in the sliding window can be summarized as follows:

$$\chi = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_k, \dots, \mathbf{x}_L]$$

$$\mathbf{x}_k = [\mathbf{p}_{b_k}^w, \mathbf{v}_{b_k}^w, \mathbf{q}_{b_k}^w, \mathbf{b}_a^{b_k}, \mathbf{b}_\omega^{b_k}, s_k], k \in [0, L] \quad (6)$$

where, L denotes the length of the sliding window of the factor graph. In time t_k , the states include: pose state: position $\mathbf{p}_{b_k}^w$ and rotation $\mathbf{q}_{b_k}^w$, velocity state $\mathbf{v}_{b_k}^w$, bias state of acceleration $\mathbf{b}_a^{b_k}$ and gyro $\mathbf{b}_\omega^{b_k}$, and scale state s_k . Unlike estimating each feature's inverse depth, we only estimate a monocular scale for an image. This monocular scale is used to convert the monocular visual relative pose measurement into a metric scale.

The error function of the proposed 4DRE-VIO can be expressed as follows:

$$\chi^* = \arg \min_{\chi} \left\{ \|\mathbf{r}_P(\chi)\|^2 + \sum_{k=1}^L \left\| \mathbf{r}_{V_{adj}}(\tilde{\mathbf{z}}_{b_k}^{b_{k-1}}, \chi) \right\|_{\Sigma_{V_{adj}}}^2 + \sum_{k=1}^L \left\| \mathbf{r}_{V_{co}}(\tilde{\mathbf{z}}_{b_{margi}}^{b_{k-7}}, \chi) \right\|_{\Sigma_{V_{co}}}^2 + \sum_{k=1}^L \left\| \mathbf{r}_{NHC}(\tilde{\mathbf{z}}_{v_k}, \chi) \right\|_{\Sigma_{NHC}}^2 + \sum_{k=1}^L \left\| \mathbf{r}_S(\tilde{\mathbf{z}}_{s_k}, \chi) \right\|_{\Sigma_S}^2 + \sum_{k=1}^L \left\| \mathbf{r}_{IMU}(\tilde{\mathbf{z}}_{b_k}^{b_{k-1}}, \chi) \right\|_{\Sigma_{IMU}}^2 \right\} \quad (7)$$

where, $\mathbf{r}_P$ represents the prior information obtained through marginalization. $\mathbf{r}_{V_{adj}}$ and $\mathbf{r}_{V_{co}}$ denote the visual adjacent factor and the optimal co-visible factor, respectively. $\mathbf{r}_{NHC}$, $\mathbf{r}_S$, and $\mathbf{r}_{IMU}$ represent NHC factor, scale prior factor, and IMU factor, respectively. Σ stands for the covariance matrix corresponding to each factor, which we assume all follow the

Gaussian distribution. The Levenberg-Marquardt algorithm in Ceres Solver [56] is used to perform non-linear optimization. In the following, we will describe each factor in detail.

A. IMU Factor

In time $[t_{k-1}, t_k]$, we use the method in VINS-Mono [28] for the construction of the IMU pre-integration residual term:

$$\mathbf{r}_{IMU}(\tilde{\mathbf{z}}_{b_k}^{b_{k-1}}, \chi) = \begin{bmatrix} \delta \alpha_{b_k}^{b_{k-1}} \\ \delta \beta_{b_k}^{b_{k-1}} \\ \delta \theta_{b_k}^{b_{k-1}} \\ \delta \mathbf{b}_a \\ \delta \mathbf{b}_\omega \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{b_k}^{b_{k-1}} \left(\mathbf{p}_{b_k}^w - \mathbf{p}_{b_{k-1}}^w + \frac{1}{2} \mathbf{g}^w \Delta t_k^2 - \mathbf{v}_{b_{k-1}}^w \Delta t_k \right) - \alpha_{b_k}^{b_{k-1}} \\ \mathbf{R}_{b_k}^{b_{k-1}} \left(\mathbf{v}_{b_k}^w + \mathbf{g}^w \Delta t_k - \mathbf{v}_{b_{k-1}}^w \right) - \beta_{b_k}^{b_{k-1}} \\ 2 \left[\left(\mathbf{q}_{b_{k-1}}^w \right)^{-1} \otimes \mathbf{q}_{b_k}^w \otimes \left(\gamma_{b_k}^{b_{k-1}} \right)^{-1} \right]_{xyz} \\ \mathbf{b}_a^{b_k} - \mathbf{b}_a^{b_{k-1}} \\ \mathbf{b}_\omega^{b_k} - \mathbf{b}_\omega^{b_{k-1}} \end{bmatrix} \quad (8)$$

where, $\delta \alpha_{b_k}^{b_{k-1}}$, $\delta \beta_{b_k}^{b_{k-1}}$, and $\delta \theta_{b_k}^{b_{k-1}}$ represent the position, velocity and rotation pre-integration residual terms, respectively. $\alpha_{b_k}^{b_{k-1}}$, $\beta_{b_k}^{b_{k-1}}$, and $\gamma_{b_k}^{b_{k-1}}$ represent the position, velocity and rotation pre-integration terms, respectively. $\mathbf{g}^w$ denotes the gravity vector in the world frame. $[\cdot]_{xyz}$ denotes the imaginary part of a quaternion.

B. Monocular Visual Odometry Factor

Each time a visual relative pose measurement is received, a state node corresponding to the timestamp is created in the factor graph. The newly added pose and scale state nodes are connected to the old nodes through monocular VO factors. Based on the visual adjacent and optimal co-visible relative pose measurement constructed in Section IV-C, the monocular VO factors can be divided into adjacent factors and optimal co-visible factors. The former connects nodes at adjacent timestamps, while the latter connects nodes at the oldest keyframe timestamp and the marginalized frame timestamp, which are the optimal associated frames with co-visible. Since both are derived from monocular visual relative pose measurements, their residual terms are similar. We model them as follows:

$$\mathbf{r}_V(\tilde{\mathbf{z}}_{b_j}^{b_i}, \chi) = \begin{bmatrix} \delta \gamma_{b_j}^{b_i} \\ \delta \eta_{b_j}^{b_i} \end{bmatrix} = \begin{bmatrix} \left[(\mathbf{q}_b^b \otimes \tilde{\mathbf{q}}_{c_j}^{c_i} \otimes \mathbf{q}_b^c)^{-1} \otimes (\mathbf{q}_{b_i}^{w-1} \otimes \mathbf{q}_{b_j}^w) \right]_{xyz} \\ s_j \mathbf{R}_{b_i}^{b_j} \left(\mathbf{R}_b^c \tilde{\mathbf{R}}_{c_j}^{c_i} \mathbf{P}_b^c + \mathbf{R}_b^c \tilde{\mathbf{P}}_{c_j}^{c_i} + \mathbf{P}_b^c \right) - (\mathbf{P}_{b_j}^w - \mathbf{P}_{b_i}^w) \end{bmatrix} \quad (9)$$

where, $\mathbf{r}_{V_{abj}}(\tilde{\mathbf{z}}_{b_k}^{b_{k-1}}, \chi)$ and $\mathbf{r}_{V_{co}}(\tilde{\mathbf{z}}_{b_{margi}}^{b_{k-7}}, \chi)$ is constructed in the same way as $\mathbf{r}_V(\tilde{\mathbf{z}}_{b_j}^{b_i}, \chi)$. $\delta\gamma_{b_j}^{b_i}$ and $\delta\eta_{b_j}^{b_i}$ represent the relative rotation error and position error, respectively. j and i refer to the associated newer state node and older state node numbers, respectively. $\otimes$ stands for quaternion multiplication. $\tilde{\mathbf{q}}_{c_j}^{c_i}$ and $\tilde{\mathbf{p}}_{c_j}^{c_i}$ are the quaternion and translation vectors corresponding to the visual relative pose measurements, respectively. $\mathbf{P}_b^c$ and $\mathbf{R}_b^c$ represent the translation and rotation components of the extrinsic parameter between the camera frame and the body frame, respectively. Note that when a scale transformation is performed on the relative pose, the rotation matrix remains unchanged, and only the translation vector is scaled according to the scale factor.

Remark: When the vehicle is driving in a straight line, the visual sliding window has a large spatial span, good co-visibility between keyframes, and many distant feature points, which result in high yaw angle accuracy. However, when the vehicle turns, especially in residential districts, too close to the object in front causes the scene to switch too quickly. In such cases, the visual sliding window has a short spatial span, limited co-visibility between keyframes, and many nearby feature points, which lead to lower yaw angle accuracy.

We propose a covariance adaptive adjustment strategy for the visual factors, as shown in (10). When the vehicle is driving in a straight line, the covariance of the yaw angle in the visual factor is reduced to fully utilize the good attitude observation provided by the visual odometry. During turning conditions, the covariance is increased to reduce its confidence. In such cases, the yaw angle is primarily estimated using the IMU pre-integration factor to take advantage of its higher short-term recursive accuracy and to achieve complementarity with the visual method. This approach mitigates the adverse effects of poor visual yaw angle estimation and significantly improves the accuracy of yaw angle estimation during vehicle turns.

$$\Delta_{TH} = (\omega_{Z,mea} > \Theta_{w_z}) \parallel (\theta_{adj} > \Theta_{adj}) \parallel (\theta_{co} > \Theta_{co})$$

$$\sigma_{yaw}^{end} = \begin{cases} \sigma_{yaw} * (1/\xi), & \text{if } \Delta_{TH} = 0 \\ \sigma_{yaw} * \xi, & \text{if } \Delta_{TH} = 1 \end{cases} \quad (10)$$

where, Δ_{TH} is the turning flag, which is triggered by three conditions. $\omega_{Z,mea}$ denotes the mean value of the angular velocity around the Z-axis measured by the IMU between adjacent nodes. θ_{adj} and θ_{co} indicates the steering angle of visual adjacent and optimal co-visible relative pose measurements, respectively. Θ_{w_z} , Θ_{adj} , and Θ_{co} denote the corresponding thresholds, respectively. σ_{yaw} and σ_{yaw}^{end} are the initial and adjusted yaw angle covariance of the visual factor, respectively. ξ is the adjustment coefficient, which is set as 10 through extensive experimental tests.

C. NHC Factor

The ego velocity information $\mathbf{v}_{ego2D}^r$ in 4D radar frame calculated in Section IV-A can provide good position and velocity constraints. Due to noise in the 4D radar point cloud caused by specular reflection, ghost reflection, and multipath

effects, etc. [39], we only utilize the estimated longitudinal velocity to mitigate the impact of noise. In addition, the lateral and vertical velocity observations are set to zero by adopting the NHC, which assumes that the vehicle does not exhibit lateral slip or vertical jump during driving. The final velocity measurement in the body frame is $\hat{\mathbf{v}}_t = [\mathbf{R}_r^b \mathbf{v}_{ego_x}^r \ 0 \ 0]$. When $t = t_k$, we can get $\hat{\mathbf{v}}_t$ as $\hat{\mathbf{v}}_k$. Within the time interval $[t_{k-1}, t_k]$, the residual term of the constructed NHC factor is:

$$\mathbf{r}_{NHC}(\tilde{\mathbf{z}}_{v_k}, \chi) = \left[\begin{pmatrix} \mathbf{P}_{b_k}^w - \mathbf{P}_{b_{k-1}}^w \\ \mathbf{v}_{b_k}^w - \mathbf{R}_{b_k}^w \hat{\mathbf{v}}_k \end{pmatrix} - \begin{pmatrix} \mathbf{R}_{b_k}^w \hat{\mathbf{v}}_k - \mathbf{R}_{b_{k-1}}^w \hat{\mathbf{v}}_{k-1} \\ \Delta t/2 \end{pmatrix} \right] \quad (11)$$

where, $\mathbf{R}_r^b$ represents the rotation matrix from the 4D radar frame to the body frame. The covariance of the NHC factor depends on the estimated longitudinal velocity variance from the 4D radar point cloud.

As shown in (11), we design the NHC factor to include both direct position constraints via velocity pre-integration and direct velocity constraints. In our application, we have found that this design is more beneficial for improving the accuracy and stability of the algorithm. Additionally, the introduction of the NHC factor not only improves horizontal positioning accuracy but also reduces vertical position drift in VIO. It is worth noting that according to the zero velocity detection results in Section IV-B, we perform zero velocity update when the vehicle is static. On the one hand, we set all velocity measurements in the NHC factor to zero. On the other hand, we only add IMU factors and NHC factors to the factor graph to achieve zero velocity update. The “Zero Velocity Update” part of Fig. 6 shows the factor graph structure at this time.

D. Scale Prior Factor

We introduce the scale prior factor to address the challenges of scale ambiguity and scale drift in monocular VO. Using the estimated ego velocity measurement $\mathbf{v}_{ego}^r$ based on the 4D radar in Section IV-A and the visual adjacent relative translation $\tilde{\mathbf{p}}_{c_k}^{c_{k-1}}$ calculated in Section IV-C, we construct the residual of the scale prior factor, which is formulated as follows:

$$\mathbf{r}_S(\tilde{\mathbf{z}}_{S_k}, \chi) = s_k - \frac{(\mathbf{R}_r^b \mathbf{v}_{ego_x,k}^r + \mathbf{R}_r^b \mathbf{v}_{ego_x,k-1}^r) / 2}{\mathbf{R}_c^b \tilde{\mathbf{p}}_{c_k}^{c_{k-1}} / (t_k - t_{k-1})} \quad (12)$$

From (12), we can see that the accuracy of the longitudinal velocity estimation determines the accuracy of the monocular scale estimation. Therefore, we set the covariance of the scale prior factor according to the variance of the longitudinal velocity estimation.

VI. EXPERIMENTAL RESULTS

In this section, to provide extensive and fair experimental tests for validating the effectiveness of the proposed 4DRE-VIO, we conduct many comparative experiments on both our dataset and a public dataset. In addition, we provide extensive ablation experiments to demonstrate the impact of each module on the results. Finally, the real-time performance of the algorithm is discussed. The experimental setup and results are presented in the following.

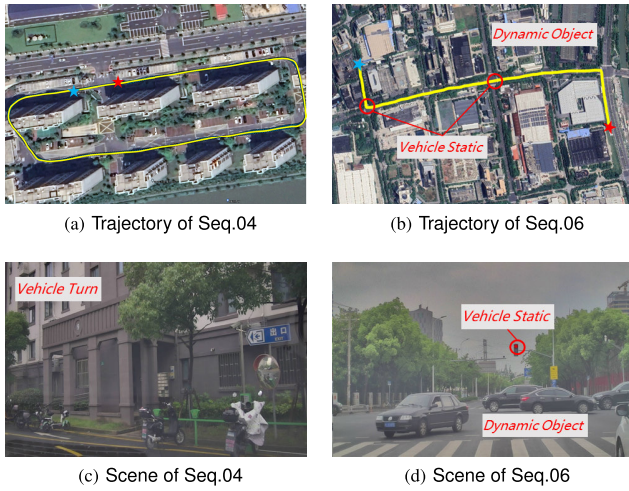


Fig. 7. Examples of trajectories and scenes in our dataset.

TABLE I

SUMMARY OF SIX SEQUENCES OF OUR DATASET

Sequence	Seq.01	Seq.02	Seq.03	Seq.04	Seq.05	Seq.06
Scene	Residential district					Open street
Time (s)	349.5	596	416	395	595	187.5
Length (m)	856	1520	879	919	1480	1759

A. Dataset

1) *Our Dataset*: We choose a Volkswagen ID4 data collection vehicle as the experimental platform, equipped with a wide-angle camera, a 4D radar, and a high-precision GNSS/IMU fusion positioning device. All devices are time-synchronized through hardware. The camera and 4D Radar operate at a frequency of 10 Hz. The used 4D Radar is GPAL Ares-R7861-4D-MRIR, which has an azimuth field of view $\pm 75^\circ$ and an elevation field of view $\pm 15^\circ$. The azimuth and elevation angle resolution are 0.7° and 0.5° , respectively. The velocity resolution is 0.2 m/s , and the detection range is up to 250 m . Each radar target point has 3D position, velocity, power and time information. The high-precision GNSS/IMU fusion positioning device provides IMU data and ground truth (GT) positioning at 100 Hz. We have collected six sequences from urban driving scenarios in Shanghai to validate the proposed algorithm, with the experimental trajectories shown in Fig. 7. Seq.01-05 are collected in residential districts. Seq.06 is an open street scenario with many dynamic objects, which pose significant challenges for positioning. In Fig. 7, the yellow line represents the ground truth trajectory of the vehicle. The blue and red stars indicate the start and end positions of the trajectory. The trajectory lengths and total duration of the six sequences are shown in Table I.

2) *Public Dataset*: The NTU4DRadLM [18] dataset is publicly released in 2023, and is specifically designed for 4D radar applications. Its introduction enriches the experimental evaluation by providing diverse urban scenarios (e.g., gardens, parking lots, and campus roads) and multi-platform data (e.g., a low-speed handcart and a high-speed car). This combination enhances the comprehensiveness and representativeness of the evaluation, effectively validating the algorithm's adaptability

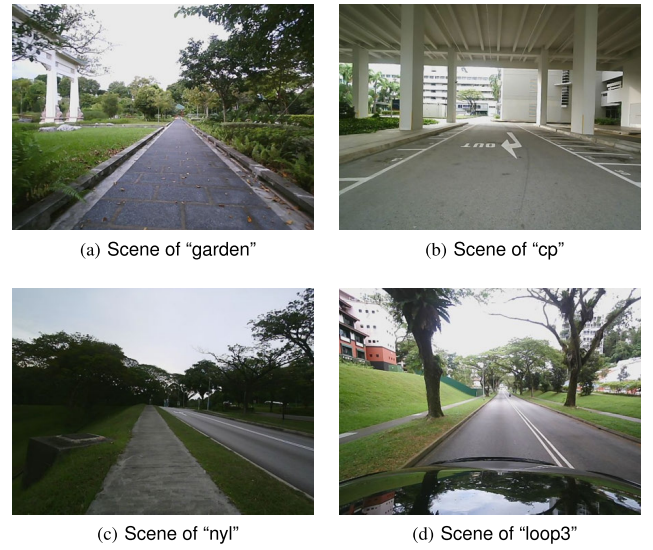


Fig. 8. Examples of scenes in the public dataset.

TABLE II

SUMMARY OF FOUR SEQUENCES OF PUBLIC DATASET

Sequence	garden	cp	nyl	loop3
Scene	Garden	Parking lot	Campus road	Campus road
Platform	Handcart			Car
Velocity	1 m/s			30 km/h

capability across different environments and conditions. To compare our method with the baseline methods, we select part of medium and large-scale sequences for experiments, including four sequences: garden, cp, nyl, and loop3, corresponding to the scenes shown in Fig. 8. The information for the four sequences is summarized in Table II.

B. Evaluation Metrics and Baselines

In order to quantitatively evaluate the performance of the proposed method, we use relative pose error (RPE). RPE evaluates the accuracy of an algorithm by measuring the difference in pose changes between the estimated pose and the ground truth pose over fixed time intervals or distance intervals, including both translation and rotation errors. Furthermore, we have used the average root mean square error (RMSE) in translation (m/m) and rotation (deg/m) for all possible sub-sequences varying in length from 20 m to 160 m , with steps of 20 m , as a comparison.

To evaluate the effectiveness of the proposed algorithm, we perform comparative experiments with seven popular algorithms on two datasets. These experiments not only validated the enhancement effect of 4D radar on VIO, but also demonstrated the advanced nature of the proposed tightly coupled framework from a multi-sensor fusion perspective. Furthermore, we analyze the performance of different algorithms in various scenes. Initially, we have performed comparisons with four state-of-the-art VIOs, including DM-VIO [34], VINS-Fusion (w/o loop) [29], VINS-Fusion-Loop (w/ loop) [29], and PL-VINS [57]. Note that we only fuse monocular vision with IMU in VINS-Fusion. This comparison

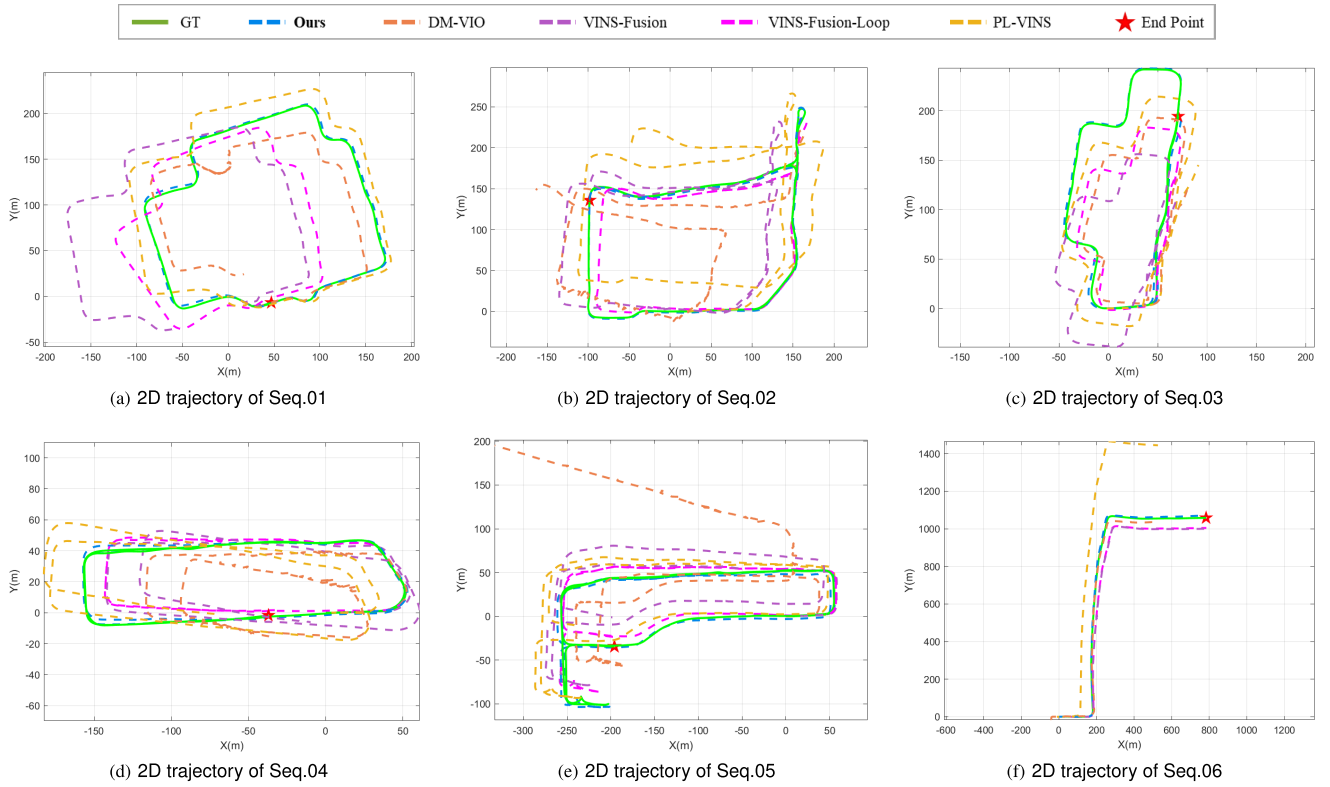


Fig. 9. 2D trajectory results of the proposed method, DM-VIO, VINS-Fusion, VINS-Fusion-Loop, and PL-VINS on six sequences.

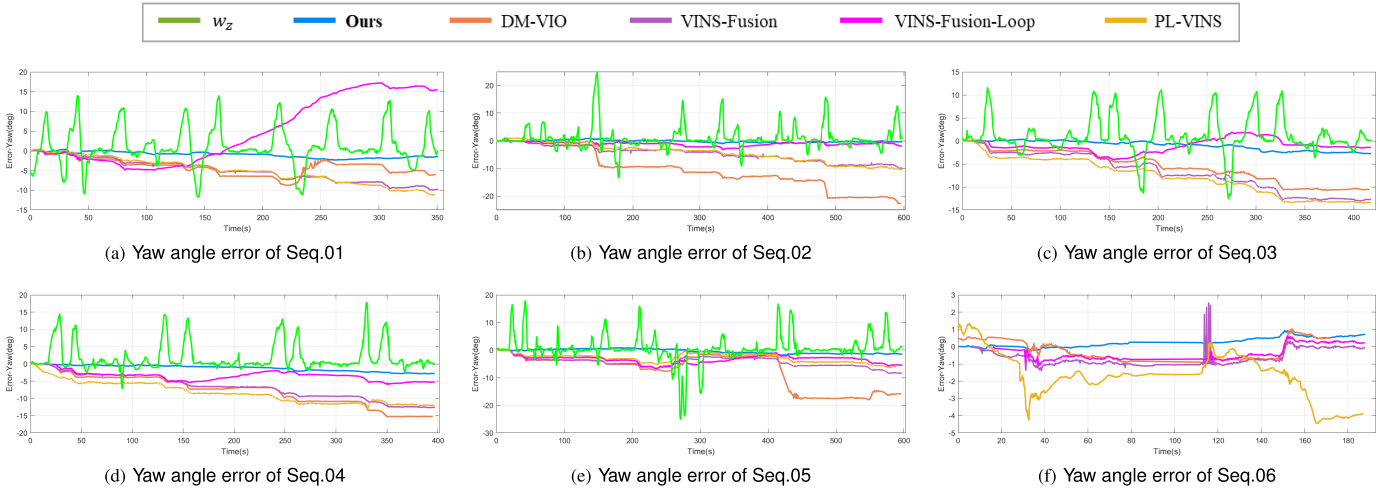


Fig. 10. Yaw angle error of the proposed method, DM-VIO, VINS-Fusion, VINS-Fusion-Loop, and PL-VINS on six sequences. In order to illustrate the vehicle driving condition: driving in a straight line or turning, we also show the angular velocity of the IMU around the Z-axis.

demonstrates the enhancement effect of the 4D radar on VIO. Additionally, to further validate the innovation of the proposed IMU-vision-4D radar tightly-coupled system, we design multi-dimensional comparative experiments on the public dataset. The experiments are categorized according to sensor combinations as follows: visual odometry (VO) DSO [33], visual-inertial odometry (VIO) VINS-Fusion [29], radar-inertial odometry (RIO) EKF-RIO [48], and radar-visual-inertial odometry (RVIO) RRxIO [50].

C. Performance Evaluation

1) *Our Dataset*: To verify the enhancement of VIO achieved by the 4D radar in the proposed algorithm, we perform comparative experiments on our dataset. Fig. 9 shows the

positioning trajectories of all evaluated algorithms across six sequences. The positioning trajectory of the proposed method almost overlaps with the GT, demonstrating high accuracy and stability even in multiple loop scenarios. In contrast, the trajectories of DM-VIO, VINS-Fusion, VINS-Fusion-Loop, and PL-VINS show scale shifts on the different sequences compared to the GT. In Seq.05, DM-VIO suffers from severe scale estimation errors in the later periods, causing its trajectory to exceed the plotting area and thus not be fully represented. This is because monocular scale estimation is inaccurate under degraded conditions due to the lack of sufficient accelerometer motion excitation. In addition, these methods show noticeable deviations from the GT due to significant yaw angle errors, as shown in Fig. 10. In the multiple loop Seq.02-05,

TABLE III
THE RPE OF THE EVALUATED ALGORITHMS ON SIX SEQUENCES (TRANSLATION ERROR t_{rel} (m/m), ROTATION ERROR r_{rel} (deg/m)).
THE BEST RESULTS ARE SHOWN IN RED AND THE SECOND BEST IN BLUE

Method	Seq.01		Seq.02		Seq.03		Seq.04		Seq.05		Seq.06		Mean	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
DM-VIO [34]	0.20362	0.02525	0.16322	0.03332	0.22206	0.0197	0.27112	0.0276	0.51803	0.03104	0.06672	0.01625	0.10328	0.02552
VINS-Fusion [29]	0.12475	0.02441	0.08829	0.02526	0.20109	0.02596	0.08648	0.02675	0.05625	0.02139	0.06283	0.02606	0.10328	0.02497
VINS-Fusion-Loop [29]	0.11929	0.04436	0.08092	0.02367	0.20508	0.02473	0.09243	0.0266	0.04421	0.02202	0.05722	0.01581	0.09986	0.02619
PL-VINS [57]	0.08732	0.02527	0.10849	0.0244	0.0892	0.0251	0.0759	0.02375	0.06737	0.02152	0.29365	0.03489	0.12032	0.02582
Ours	0.01744	0.01629	0.01839	0.01995	0.01804	0.01613	0.01961	0.01815	0.02149	0.01642	0.01477	0.01354	0.01829	0.01675

TABLE IV
COMPARISON OF RPE OF EVALUATED ALGORITHMS ON NTU4DRADLM DATASET (TRANSLATION ERROR t_{rel} (m/m),
ROTATION ERROR r_{rel} (deg/m)). THE BEST RESULTS ARE SHOWN IN RED AND THE SECOND BEST IN BLUE

	Method	cp		garden		nyl		loop3		Mean		Improvement	
		t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
VO	DSO [33]	0.71371	0.03790	0.66071	0.03583	0.90979	0.02236	0.96826	0.01396	0.81312	0.02751	97.13%	11.74%
VIO	VINS-Fusion [29]	0.12646	0.03090	0.04067	0.03646	0.34184	0.02124	0.05392	0.01350	0.14072	0.02553	83.42%	4.9%
RIO	EKF-RIO [48]	0.11722	0.05486	0.24218	0.49125	0.11628	0.12690	0.25767	0.05806	0.18334	0.18277	87.28%	86.72%
RVIO	RRxIO [50]	0.21187	0.10023	0.13581	0.16703	0.55824	0.08990	×	×	0.22648	0.11905	89.70%	79.61%
	Ours	0.02135	0.03053	0.02022	0.03573	0.02051	0.02177	0.03123	0.00909	0.02333	0.02428	/	/

this problem is more obvious. Although VINS-Fusion-Loop ensures the consistency of multiple loop trajectories through loop closure function, it still suffers from inherent monocular scale and yaw angle errors. The proposed method makes full use of the metric scale information provided by the 4D radar, and significantly improves the monocular scale recovery, which is strongly contrasted by the basic DM-VIO.

Furthermore, as shown in Fig. 10, apart from our method, VIO methods typically exhibit large yaw errors during vehicle turns, with this problem being more pronounced in residential districts (Seq.01-05). By analyzing the turning scene shown in Fig. 7c, it is clear that this is caused by rapid scene switches and limited visual co-visible regions during near-field turns. Our algorithm addresses this limitation by adaptively adjusting the covariance of visual factors and fully leveraging IMU attitude prediction, thereby significantly improving yaw angle estimation accuracy during turns.

Table III shows the quantitative comparison results between the proposed method and the baseline methods. In terms of relative translation error and relative rotation error metrics on six sequences, our method demonstrates superior performance compared to other methods. Specifically, our method outperforms the second best method by 81.68% in the average relative translation error and by 32.92% in the average relative rotation error. This confirms the high level of accuracy and stability of the 4DRE-VIO that we have proposed in urban environments. In particular, extensive comparisons with VIO algorithms confirm that the tight integration of 4D radar with monocular VIO significantly enhances performance.

2) *Public Dataset*: To validate the effectiveness of the proposed tightly-coupled framework integrating 4D radar, vision, and IMU from a multi-sensor fusion perspective, we perform multi-dimensional comparative experiments on the

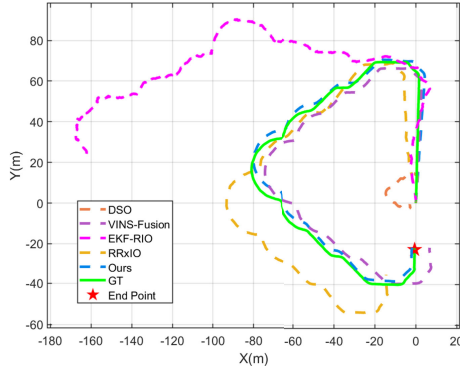
public dataset. Table IV presents the quantitative performance comparison with baseline methods, demonstrating that our method achieves the better performance. Specifically, as shown in Fig. 8, the garden, parking lot, and campus road scenes exhibit rich visual features, enabling DSO to achieve high rotation estimation accuracy. However, monocular scale ambiguity and drift result in poor translation accuracy. VINS-Fusion shows similar rotation accuracy but improves translation accuracy through IMU integration, though it remains limited in low-speed handcart scenes (e.g., cp and nyl) due to scale unobservability under degenerate motion. In contrast, our algorithm leverages 4D radar for precise scale recovery, significantly enhancing translation accuracy. Additionally, the adaptive covariance adjustment strategy for visual factors improves rotation accuracy compared to VO and VIO.

Compared to the RIO algorithm, our method improves relative translation and rotation accuracy by 87.28% and 86.72%, respectively. This is because EKF-RIO relies solely on vehicle velocity as an observation, leading to low rotation observability and accuracy. While IMU bias causes error accumulation, worsened in low-speed scenes due to poor signal-to-noise ratio. This is especially evident in the garden scene with frequent turns. Rotation errors further degrade translation accuracy through integration, particularly in high-speed scenes like loop3, where minor errors cause large deviations. Our method benefits from visual observations' rotation constraints, highlighting the complementary role of visual information to RIO.

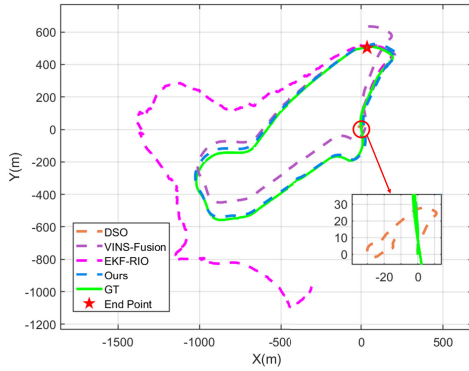
Furthermore, compared to the RVIO algorithm, our method also shows significant improvement. Although RRxIO integrates three sensors, its loosely coupled architecture limits the full potential of visual and 4D radar data integration. This means that the reduction in VIO or RO accuracy will result in

TABLE V
COMPARISON OF RPE FOR THE QUANTITATIVE RESULTS OF THE ABLATION STUDY ON SEQ.04 (TRANSLATION ERROR t_{rel} (m/m), ROTATION ERROR r_{rel} (deg/m)). THE BEST RESULTS ARE SHOWN IN RED

Method	First		Second		Mean	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
Ours (w/o Adaptive Covariance Adjustment Strategy)	0.02503	0.02787	0.02501	0.02785	0.02502	0.02786
Ours (w/o the Velocity Measurement from 4D Radar)	0.38620	0.07230	0.31506	0.06148	0.35063	0.06689
Ours (Full)	0.01960	0.01874	0.01959	0.01851	0.019595	0.01862



(a) 2D trajectory of "garden"



(b) 2D trajectory of "loop3"

Fig. 11. Examples of 2D trajectory on NTU4DRadLM dataset ("garden" and "loop3").

a reduction in final fusion accuracy. As analyzed earlier, in the three low-speed scenes, VIO suffers from poor translation accuracy due to monocular scale ambiguity. Direct fusion with RO may even result in lower accuracy than using RIO alone. Even in the loop3, RRxIO fails entirely due to VIO's initialization failure. By tightly coupling vision, 4D radar, and IMU information based on an optimization-based approach, our algorithm fully leverages their complementary strengths and thus performs significantly better than the loosely coupled RRxIO.

The qualitative results are shown in Fig. 11, which presents the experimental results of some sequences. It can be seen that the positioning trajectory of our method is closest to the GT, further confirming the effectiveness of the proposed 4D radar-enhanced monocular visual inertial odometry method.

D. Ablation Study

To validate the contribution of each module to the proposed method, we conduct an ablation study on our dataset Seq.04

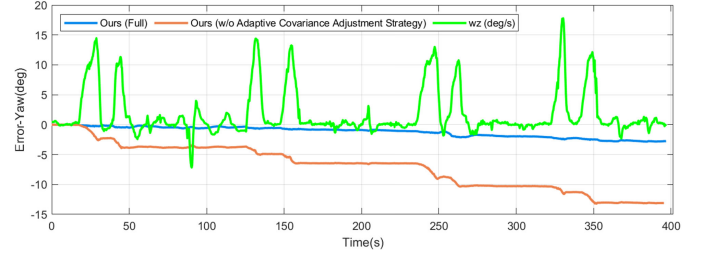


Fig. 12. The results of yaw angle error for the Seq.04.

with two tests. The ablation modules and the quantitative experimental results are summarized in Table V. In addition, to test the effectiveness of the dynamic object removal module, we perform an ablation study on open street Seq.06, which contains many dynamic objects.

1) *Benefits of Adaptive Covariance Adjustment Strategy:* The adaptive covariance adjustment strategy adjusts the weight of yaw angle in visual factors according to the driving condition. This module is useful for improving the accuracy of attitude estimation, particularly important when turning when close to the object. The results in Table V show that after ablating this module, the positioning accuracy of the proposed method decreases, especially in terms of rotation accuracy. Fig. 12 illustrates the yaw angle estimation error, where we can observe significant errors during the turning condition after removing this module. However, introducing the adaptive covariance adjustment strategy for visual factors significantly improves this issue. This is due to the beneficial support provided by the rotation constraints from the IMU at this time.

2) *Effect of the Velocity Measurement From 4D Radar:* The factors associated with 4D radar velocity measurement include the NHC factor and the scale prior factor. Ablating this module means that these two factors are not included in the factor graph optimization, resulting in the algorithm degrading to a tightly coupled visual and IMU odometry. The results in Table V indicate that without the constraints of 4D radar velocity measurement, the accuracy of the odometry decreases significantly. Fig. 13 shows the results of scale estimation under these conditions. It can be seen that after the removal of this module, the scale estimation accuracy decreases significantly, which is the main reason for the decrease in positioning accuracy. In addition, due to the presence of many turning and even fast turning scenarios in Seq.04, the visual point depth estimation could not converge fast enough, resulting in a large error, which leads to a large range of scale variations. Nevertheless, the introduction of 4D radar velocity measurements provides metric scale information

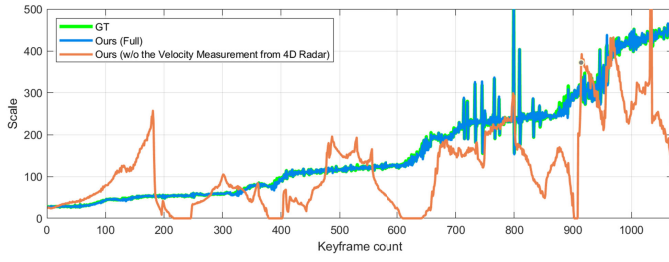


Fig. 13. Results of the monocular scale estimation for the Seq.04. The ground true of scale is calculated using the relative translation measurements between adjacent keyframes from the ground truth trajectory and our algorithm estimated relative translation measurements. “Ours (Full)” represents the scale estimation results of our algorithm. It can be seen that the monocular scale is time-varying, and our algorithm achieves impressive scale estimation accuracy.

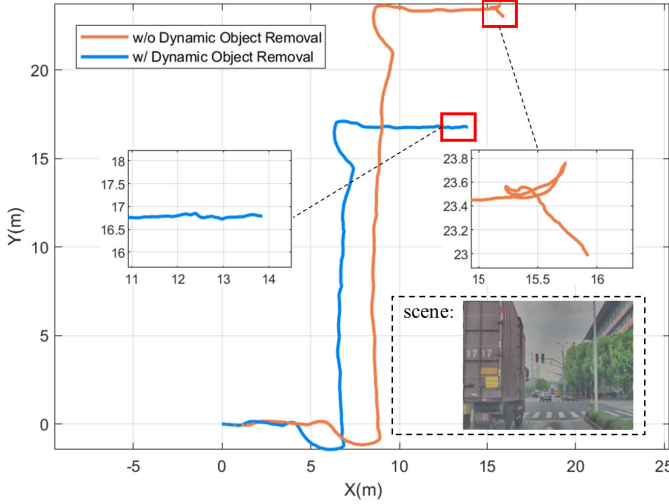


Fig. 14. Positioning results of monocular VO with or without the dynamic object rejection module.

that allows good estimation and recovery of monocular scales. The close alignment of the Ours (Full) with the GT in Fig. 13 also confirms this. In addition, it can be seen from Table V that the removal of this module also significantly increases the rotation error. This is mainly due to the fact that relying solely on monocular VO relative pose observations cannot effectively constrain and correct the six-axis bias of IMU, especially when crossing bumps. This leads to a significant increase in the rotation error of the IMU pre-integration afterwards.

3) *Effect of the Dynamic Object Removal*: By influencing the visual positioning result, the dynamic object removal module affects the final positioning accuracy. Therefore, to better illustrate the role of this module, we only compare the results of monocular VO (without scale recovery) in this study. Positioning results with or without dynamic object removal for the monocular VO module proposed in this paper are shown in Fig. 14. It is evident that our algorithm achieves more accurate pose estimation through precise dynamic object removal. Specifically, in the moving truck scene (red boxed area) described in Section IV-C1, the algorithm without dynamic object removal incorrectly estimates the motion of the vehicle as backing up. This is mainly because the velocity of the truck is higher than that of the vehicle, and the high-gradient points extracted at this time are mainly distributed on the truck, as shown in Fig. 2b. Our algorithm can prevent dynamic points

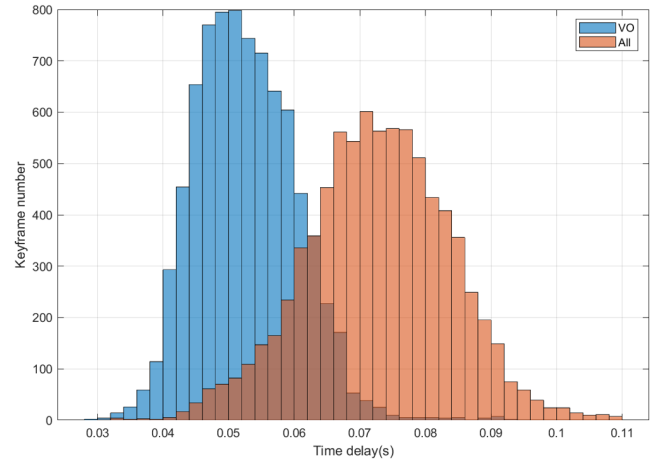


Fig. 15. Histogram of the frequency distribution of the positioning time delay.

from being involved in pose estimation from the root, which effectively solves this problem. As can be seen in Fig. 4c, the high-gradient points extracted at this time are no longer distributed on dynamic objects.

E. Computation Time

We evaluate the real-time performance of the system by measuring the time delay of positioning in six sequences. The time delay refers to the difference between the time of obtaining the output positioning result and the image. Due to the performance limitations of the computing platform (GeForce RTX 2060), we do not count the time consumption of the InternImage [54] (A100 GPU). The histogram of the frequency distribution of the time delay is shown in Fig. 15. Among them, the average total positioning time delay for a single keyframe is 72.77 ms, the average for VO is 52.67 ms, and the average for factor graph optimization is 20.10 ms. As a result, the time delay for the positioning of the proposed algorithm is relatively low. A higher rate of state output can be obtained using the IMU pre-integration.

VII. CONCLUSION

To the best of our knowledge, this study is the first positioning system that tightly integrates the 4D radar, IMU, and monocular camera based on the factor graph. This is a novel 4D radar-enhanced monocular visual inertial odometry. In our approach, a reliable velocity observation is proposed by combining the velocity of the 4D radar with NHC. Meanwhile, a direct constraint on the monocular scale is constructed using the velocity, which ensures a reliable estimation of the monocular scale. A dynamic object removal algorithm based on 4D radar-enhanced semantic segmentation is proposed to mitigate the interference from dynamic objects. An adaptive fusion strategy based on driving conditions is proposed, including adaptive adjustment of visual measurement covariance in turning conditions and adaptive adjustment of fusion information in static conditions. This fully utilizes the advantages of each sensor, significantly improving positioning accuracy, especially the accuracy of yaw angle estimation. Extensive real-world experiments and ablation studies in urban

environments confirm the high level of accuracy and stability of the proposed method. Notably, compared to SOTA in VIO, RIO and RVIO, our method shows superior performance on our dataset and the public dataset.

In the future, we will fully leverage the advantages of 4D radar in adverse weather conditions to further explore the performance of the proposed method under degraded conditions. Additionally, we plan to incorporate more urban scenes and non-flat terrain data to comprehensively validate and refine the proposed algorithm. Finally, we aim to integrate deep learning techniques to achieve online adaptive adjustment of covariance parameters. We believe this improvement holds significant potential for the field of multi-sensor fusion.

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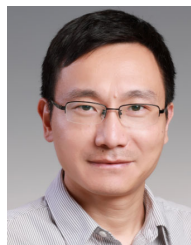
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